Project Name - DegreePath

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Group - Software Engineering I I2 - 1

Instructor - Raz Sarem

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AI disclosure - GPT-5 used for proofreading

Personas

P1- Students(Primary): An NYU undergraduate CS major in their 2nd or 3rd year planning their remaining semesters. They want a valid plan under prerequisite, credit-cap, and term-offering constraints, and an explanation when no plan works.

P2- Department Catalog Administrator: Maintains course data and degree requirements. They need bad data rejected before it reaches the planner. In the MVP this is a seeded admin account using file import, with the in-app editor deferred to the MLP.

P3- Persona change (stated reason): *Academic Advisor* is removed from the MVP persona list. The approved MVP has no advisor-facing feature (the advisor view is an MLP item), and students share their plan with advisors via CSV. Advisor stories return in a later sprint.

User story backlog

Dependency map: US-01, US-05 → US-02; US-05 → US-04, US-06, US-08; US-06 → US-07; US-02, US-06, US-08 → US-09; US-09 → US-10, US-12; US-08, US-09 → US-18; US-02, US-10 → US-03; US-10 → US-11, US-13. No MVP story depends on a Stretch story.

Group A: Account and profile (Student)

US-01 | Student | MVP

As a student, I want to register and sign in with email and password, so that my profile and plans are saved privately to me.

- AC1: Given I have no account, when I register with an unused email and a password of 8+ characters, then my account is created and I land on an empty profile page.
- AC2: Given I have an account, when I sign in with a wrong password, then I see "Incorrect email or password" and no session is created.
- AC3: Given I register with an email that already exists, then I see "Email already registered" and no second account is created.
- AC4: Given I am signed out, when I open a profile or plan URL directly, then I am redirected to the sign-in page.
- Author: Maitri | Reviewer: Han | Status: Agreed | Depends on: none

US-02 | Student | MVP
As a student, I want to enter my completed courses, declare major, per-term credit limit, and target graduation term, so that the plan starts from my actual situation.

- AC1: Given I am signed in and the catalog is loaded, when I save completed courses, major, a credit limit of 15, and a target of Spring 2028, then after signing out and back in the same values are shown.
- AC2: Given I enter a completed-course code that is not in the catalog, when I save, then the save is rejected with a message naming the code and my profile is unchanged.
- AC3: Given I enter a credit limit of 0 or 30 (valid range is a whole number from 1 to 24), or a target term in the past, when I save, then a field-level error is shown and nothing is saved.
- Author: Maitri | Reviewer: Hanyue | Status: Agreed | Depends on: US-01, US-05

US-03 | Student | MVP
As a student, I want to update my profile after a term (new completed courses, changed credit limit), so that my next plan reflects where I am now.

- AC1: Given a saved profile, when I add a completed course and save, then the profile shows the new course and an "updated" timestamp.
- AC2: Given I have a saved plan, when I change and save my completed courses, declare major, per-term credit limit, or target graduation term and then open the plan, then it is labeled “Out of date — regenerate.”
- AC3: Given another student's profile ID, when I send an update request for it, then the system returns 403 and that profile is unchanged.
- AC4: Given I edit my credit limit, when I save, then values 1 and 24 are accepted, and values 0 and 25 are rejected with an error stating the valid range is 1-24, and my saved profile is unchanged.
- Author: Han | Reviewer: Omkar | Status: Agreed | Depends on: US-02, US-10

US-04 | Student | MVP
As a student, I want to search the catalog by course code or title and see credits, prerequisites, corequisites, and terms offered, so that I can pick my completed courses accurately and understand a course's constraints.

- AC1: Given the catalog is loaded, when I search "algorithms," then I see matching courses with credits, prerequisites written as e.g. "A AND (B OR C)," corequisites, and terms offered.
- AC2: Given my search matches nothing, then I see "No courses found" and no error page.
- AC3: Given I am not signed in, when I send an API request to access the catalog, then the API returns HTTP 401 Unauthorized and does not return catalog data.
- AC4: Given I am not signed in, when I access the catalog through the web page, then I am redirected to the sign-in page.
- Author: Hanyue | Reviewer: Achyuthan | Status: Agreed | Depends on: US-05

Group B: Catalog management (Catalog Administrator)

US-05 | Catalog Administrator | MVP
As a catalog administrator, I want to import a CSV/JSON file of courses with credits and term offerings, so that the planner uses current department data.

- AC1: Given I am signed in as admin and upload a valid file of 3 courses, then I see "3 courses imported" and all 3 appear in the catalog.
- AC2: Given a row has missing or non-numeric credits, when I upload, then nothing is imported and the error lists the row number and column name.
- AC3: Given a course code already exists, when I import a row with updated credits or term offerings, then those fields are updated on the existing course and no duplicate is created.
- AC4: Given I am signed in as a student, when I upload a catalog file, then the system returns 403 and the catalog is unchanged.
- AC5: Given the file contains an invalid term-offering value, when I upload it, then nothing is imported and the error identifies the row number and invalid value.
- Author: Han | Reviewer: Maitri | Status: Agreed | Depends on: none

US-06 | Catalog Administrator | MVP
As a catalog administrator, I want to import AND/OR prerequisite groups and corequisites, so that course rules are modeled correctly.

- AC1: Given a file with prerequisite "A AND (B OR C)," when imported, then the course shows A as required and B or C as alternatives.
- AC2: Given courses X and Y list each other as corequisites, when imported, then the import is accepted with no cycle error.
- AC3: Given a prerequisite written as prose such as "instructor permission," when imported, then the course is imported, flagged "manual review" in the import report, and not treated as a satisfied prerequisite.
- AC4: Given a prerequisite refers to a course code not in the catalog, then the import is rejected and the message names the unknown code.
- Author: Omkar | Reviewer: Hanyue | Status: Agreed | Depends on: US-05

US-07 | Catalog Administrator | MVP
As a catalog administrator, I want imports with prerequisite cycles rejected with the cycle shown, so that bad data never reaches the planner.

- AC1: Given A requires B, B requires C, and C requires A, when I import, then the import is rejected and the message shows "A → B → C → A."
- AC2: Given a rejected import, then the catalog afterward is identical to before (same course count and rules).
- AC3: Given courses X and Y list each other as corequisites and no prerequisite cycle exists, when I import, then the import succeeds; corequisite relationships are not treated as prerequisite cycles.

- AC4: Given A requires B, B requires C, and C is a corequisite of A, when I import, then the import is rejected and the message shows the loop "A → B → C ⇄ A," because corequisites must share a term and the prerequisite chain forces them apart.
- Author: Achyuthan | Reviewer: Omkar | Status: Agreed | Depends on: US-06

US-08 | Catalog Administrator | MVP
As a catalog administrator, I want to define degree requirements for the pilot major (required courses, "choose N from" pools, credit thresholds), so that plans are checked against real graduation rules.

- AC1: Given I upload a definition with 8 required courses, a pool "choose 3 from 6," and "at least 12 credits at 3000-level or above," then a summary page shows all three rule types.
- AC2: Given a pool says "choose 5 from 4 courses," then the upload is rejected with a message naming the pool.
- AC3: Given a requirement references a course not in the catalog, then the upload is rejected and the message names the course.
- AC4: Given I am signed in as a student, when I upload requirements, then the system returns 403.
- Author: Hanyue | Reviewer: Han | Status: Agreed | Depends on: US-05

Group C: Planning (Student)

US-09 | Student | MVP
As a student, I want to generate a semester-by-semester plan, so that I know which courses to take each term.

- AC1: Given a saved profile and a valid catalog, when I click Generate, then I see courses by term from the next term through my target term, and the plan passes the independent validator: no term exceeds my credit limit, every prerequisite is completed or in an earlier term, corequisites share a term, every course is in a term where it is offered. Degree-requirement satisfaction is verified separately in US-18.
- AC2: Given I have completed course X, then X does not appear in the plan.
- AC3: Given my profile is missing a credit limit or target term, when I click Generate, then generation is blocked with a message listing the missing fields.
- AC4: Given the same profile and catalog, when I generate twice, then both plans are identical.
- AC5: Given no plan can satisfy my profile's constraints, when I click Generate, then no plan is shown, the result reads "No valid plan exists for these constraints," and the conflict explanation from US-12 is available.
- Author: Achyuthan | Reviewer: Maitri | Status: Agreed| Depends on: US-02, US-06, US-08

US-10 | Student | MVP

As a student, I want to save a plan and reopen it later, so that I don't have to regenerate it every session.

- AC1**:** Given a generated plan, when I save it, then it appears in "My plans" with a name and creation date, and is identical after sign-out and sign-in.
- AC2**:** Given I have no saved plans, then "My plans" shows "No plans yet" and a Generate button.
- AC3**:** Given a plan belonging to another student, when I request it, then the system returns 403 and the plan is not disclosed.
- Author: Omkar | Reviewer: Achyuthan | Status: Agreed | Depends on: US-09

US-11 | Student | MVP
As a student, I want to delete a saved plan, so that my list contains only plans I still care about.

- AC1: Given a saved plan, when I delete it and confirm, then it is permanently deleted, no longer appears in "My plans" and opening its URL returns 404.
- AC2: Given I click Delete and then Cancel, then the plan remains.
- AC3: Given another student's plan ID, when I send a delete request, then the system returns 403 and the plan remains.
- Author: Hanyue | Reviewer: Omkar | Status: Agreed | Depends on: US-10

US-12 | Student | MVP
As a student, when my goal is impossible, I want to see which specific constraints conflict, so that I know what to change.

- AC1: Given my target graduation leaves 4 terms but a prerequisite chain needs 5, when I generate, then no plan is shown as valid and the screen lists only the constraints that contribute to the conflict (here, the prerequisite chain and the graduation term), with no unrelated constraints listed.
- AC2: Given any reported conflict set in a test fixture, when any single listed constraint is removed, then the remaining constraints are feasible (minimality).
- AC3: Given 15–20 hand-built infeasible scenarios, then each reported conflict set is minimal (removing any one constraint makes the scenario feasible) and contains only constraints declared as conflicting in that scenario's fixture.
- AC4: Given the solver time limit is reached, then the result is labeled "possibly non-minimal."
- Author: Achyuthan | Reviewer: Han | Status: Agreed | Depends on: US-09

US-13 | Student | MVP
As a student, I want to export my plan as CSV, so that I can share it with my advisor.

- AC1: Given a saved valid plan, when I click Export CSV, then a file downloads with header `term,course_code,title,credits` and one row per scheduled course, ordered by term.

- AC2: Given an exported file, when it is fed to the validator together with the student's profile, then it passes with no errors.
- AC3: Given there is no plan or the plan is infeasible, then Export is disabled with the message "Generate a valid plan first."
- AC4: Given another student's plan ID, when I request export, then the system returns 403.
- Author: Maitri | Reviewer: Achyuthan | Status: Agreed | Depends on: US-10

US-18 | Student | MVP

As a student, I want my generated plan to satisfy every rule of my major, so that following it actually lets me graduate.

- AC1: Given the pilot major requires 8 courses, 3 from a pool of 6, and 12 credits at 3000-level or above, when I generate a plan, then the plan contains all 8 required courses, at least 3 pool courses, and at least 12 qualifying credits.
- AC2: Given a course belongs to two pools, when it appears in my plan, then it counts toward only one pool and the plan view states which.
- AC3: Given no plan can satisfy the requirements by my target term, when I generate, then no plan is shown and the conflict list from US-12 names the unsatisfied requirement.
- Author: Achyuthan | Reviewer: Han | Status: Agreed | Depends on: US-08, US-09

Stretch (in priority order)

US-14 | Student | Stretch #1

As a student, I want to run a what-if scenario by dropping a course or moving my graduation term and compare it with my current plan, so that I can see the impact before committing to the change.

- AC1**:** Given a saved plan, when I change my graduation from Spring 2028 to Fall 2028, then a side-by-side view shows both plans and highlights differing courses.
- AC2**:** Given a saved plan, when I drop a course in a what-if scenario, then the dropped course is absent from the what-if plan and the side-by-side view highlights the differences between the saved plan and the what-if plan.
- AC3**:** Given a saved plan, when I run a feasible what-if scenario, then my saved plan remains unchanged and the what-if changes are applied to my saved plan only if I explicitly choose to save the what-if plan.
- AC4**:** Given a saved plan, when I run an infeasible what-if scenario, then my saved plan remains unchanged and the conflict list from US-12 is shown.
- Author: Omkar | Reviewer: Maitri | Status: Agreed | Depends on: US-09, US-12

US-15 | Student | Stretch #2
As a student, I want plain-language suggested fixes for an infeasible plan (raise the cap, extend graduation), so that I can fix it without guessing.

- AC1: Given an infeasible result, then at least one suggestion appears, and applying it produces a feasible plan.
- AC2: Given no single change can fix it, then the message says so and lists the conflicts only.
- Author: Han | Reviewer: Hanyue | Status: Agreed | Depends on: US-12

US-16 | Student | Stretch #3
As a student, I want to export my plan and registration deadlines as an .ics file, so that I can add them to my calendar.

- AC1: Given a saved plan, when I export .ics, then the file imports into a calendar app with one event per term start.
- AC2: Given no registration dates are in the term calendar, then only term events are exported and a notice says so.
- Author: Hanyue | Reviewer: Maitri | Status: In review | Depends on: US-13

US-17 | Student | Stretch #4
As a student, I want to declare a minor alongside my major, so that my plan covers both.

- AC1: Given a minor with requirements loaded, when I add it, then the generated plan satisfies both sets of requirements.
- AC2: Given the minor is infeasible with my target term, then the conflict list names the minor's constraints.
- Author: Maitri | Reviewer: Omkar | Status: Agreed | Depends on: US-08, US-09

Non-Functional Requirements

- **NFR-1 Permissions:** A student cannot read, edit, or delete another student's profile or plans, and a non-admin cannot import catalog or requirement files. Verified by API tests returning 403 for each endpoint.
- **NFR-2 Security:** Passwords are stored only as salted hashes (bcrypt or argon2). A database inspection test finds no plaintext passwords. Sessions expire after 24 hours, and an expired token returns 401.
- **NFR-3 Performance:** Plan generation for the pilot major completes in ≤10 seconds at the 95th percentile over 20 runs. An infeasibility explanation returns within 30 seconds or is labeled "possibly non-minimal."

- **NFR-4 Data integrity:** Catalog import is all-or-nothing. A test with an invalid file shows row counts unchanged. 100% of generated plans in the automated suite (≥50 scenarios) pass the independent validator.
- **NFR-5 Privacy:** Application logs contain no completed-course lists or passwords, checked by a log-scan test on a sample run.
- **NFR-6 Persistence:** Profiles and saved plans survive a server restart on the deployed demo, verified by restart-and-compare.

Out-of-Scope List

Carried forward from Project Ideas:

- Registering for classes or integrating with NYU's registration system.
- Real-time seat availability, waitlists, or enrollment caps.
- Grade prediction or course recommendations by difficulty or instructor.
- Advising exceptions, petitions, substitutions, and "instructor permission" prerequisites (flagged for manual review, not modeled).
- Departments other than the one pilot major.

Updated this sprint (deferred beyond MVP):

- Advisor view and comments (persona removed from MVP).
- Drag-and-drop plan editor, PDF export, in-app catalog editor, and term-calendar editor.
- Minimum-grade prerequisites (listed once, Stretch/Nice-to-Have only; the MVP models course completion, not grades).
- Double majors, optimization goals, section-level scheduling, transfer/AP credit, university SSO, shareable links, and the visual graph explorer.

Scope clarification: "either/or" (OR) prerequisite groups are **MVP** (US-06). They are not a Nice-to-Have item.

Conditions Response

Condition 1: "Fix inconsistent feature lists (either/or prerequisites in MVP and nice-to-have; minimum-grade prerequisites listed twice)." → Either/or (AND/OR) prerequisite groups are MVP only, covered by US-06 AC1. Minimum-grade prerequisites are deferred; the MVP models course completion, not grades.

Condition 2: The class search page opens without login, but only pre-college courses were visible. Current-term CS-UY offerings could not be confirmed (Spring 2027 may not be published yet). Term offerings are loaded through the import file (US-05 AC5), so the MVP does not depend on the class search being readable. The manual-entry fallback from Project Ideas applies.

Review Log

3 Oct 2026 (WhatsApp, all members): Omkar opened the sprint; agreed all members edit the doc directly so version history shows contributions.

4 Oct 2026 (WhatsApp): Han requested a first draft to distribute stories. Achyuthan posted the initial structure; stories divided across members.

5 Oct 2026 (G-meet): Attendees: [Achyuthan , Omkar , Maitri , Han Yang , Hanyue]. The team walked through every open review comment and resolved them individually: US-04 split its auth criterion into API (401) and web redirect cases; US-10 now returns 403 only, consistent with NFR-1; US-13's validator check now includes the student's profile; US-07 clarified that corequisites are not prerequisite cycles and added a prereq–coreq contradiction case; US-09 gained an AC for the no-valid-plan case; US-12 now lists only constraints that contribute to a conflict; US-05, US-11 and US-14 were tightened per reviewer suggestions; and credit-limit boundaries are now tested at 1/24 and 0/25. Two disagreements were settled: degree requirements stay in US-18 rather than US-09 AC1 so US-09 remains sprint-sized, and US-15 suggestions are limited to single-change fixes. All MVP stories were marked Agreed, but 1 US-16 which is under review and needs to be discussed further.